

Shinhaeng Lee

Atlanta, GA | shinhaenglee.com | slee3487@gatech.edu | 470-676-4493

Education

Georgia Institute of Technology — Atlanta, GA

M.S. in Computer Science, Specialization in Machine Learning, GPA: 4.0 / 4.0

B.S. in Computer Science, GPA: 3.7 / 4.0

Expected Dec. 2026

Dec. 2025

Research Experience

Research Intern — OntactHealth

Medical Robotics and Echocardiography

May 2026–Aug. 2026

- Developed action-conditioned flow-matching and autoregressive video-Transformer world models that predict PLAX/A4C echocardiography from probe tilt/rock commands, using motion-aware cross-attention, adaptive normalization, temporal memory, and long-horizon rollout training.
- Integrated a frozen cardiac-segmentation teacher with differentiable Dice and cross-entropy losses to preserve anatomy and action-induced structural motion.
- Engineered a patient-disjoint pipeline for 4,994 clips (~721K frames) with balanced sampling, memory-mapped NPY loading, segmentation caching, leakage controls, and a CUDA inference dashboard for checkpoint, rollout, and anatomy-overlay comparison.
- Built ROS 2 pipelines for vision-guided anatomical localization and motion planning on the Franka Research 3 robotic arm.
- Implemented and tuned impedance/admittance controllers for compliant, stable contact during robot-assisted ultrasound acquisition.

Research Assistant — The Teachable AI Lab, Georgia Institute of Technology

Cognitive Modeling and Procedural Learning

Aug. 2025–Present

- Developed a structure-mapping method over typed abstract syntax trees, using source–target correspondences and anchored relational search to generate self-explanations of mathematical procedures.
- Evaluated the method on 70 transformations spanning 120 changed numeric components, recovering every intended procedure while reducing candidate combinations by 93.8% versus unguided search.
- Showed that structure-guided search makes procedure learning efficient and inspectable rather than relying on opaque data-driven inference, while formulating a testable cognitive hypothesis that human learners prioritize corresponding, structurally local evidence during self-explanation; first-authored the ACS 2026 submission.

Research Assistant — Sarker Lab, Emory University

Natural Language Processing for Public Health

May 2025–Jan. 2026

- Built modular substance-use extraction pipelines combining regex/lexicons, pretrained language models, and large language models for noisy social-media text.
- Analyzed longitudinal polysubstance transitions for ~112K Reddit users with Hidden Markov models, multi-order Markov chains, Apriori, and PrefixSpan sequence mining.
- Contributed to an accepted AMIA 2026 poster and the related journal manuscript currently in preparation.

Publications & Presentations

Lee, S., MacLellan, C. J., & Weitekamp, D. "Structure-Mapping-Guided Self-Explanation for Learning Mathematical Procedures." Submitted to *ACS 2026*; under review.

Das, S., Lee, S., Peddireddy, S., Lakamana, S., & Sarker, A. "Longitudinal Polysubstance Use Transitions on Reddit: A Large-scale Study using NLP" Accepted poster, *AMIA 2026 Annual Symposium*, Poster 308. Related working paper in preparation for journal submission.

Selected Projects

Dynamic Compute-Skipping for Diffusion LLM Inference

Diffusion LLMs, Dynamic Inference, Model Acceleration

Feb. 2026

- Implemented token- and layer-level skipping in Fast-dLLM v2 using cosine-similarity gating and top-*k* change detection, reducing FLOPs by 50% while maintaining GSM8K accuracy.

Real-Time YOLOv3 Detection for Gastrointestinal Endoscopy

PyTorch, OpenCV

Aug. 2024

- Fine-tuned YOLOv3 on Kvasir to mAP@0.5 of 0.74; reduced parameters 92% with L1 + Taylor pruning at ~11% accuracy loss, exceeding L1-only reduction by 20%.
- Applied *k*-means anchor optimization for 40% faster convergence and achieved 19 FPS / 54 ms latency on CPU.

Autonomous-Driving Simulation with Genetic Algorithm

Genetic Algorithms, JavaScript, HTML/CSS

Dec. 2023

- Built a browser-based driving simulator and optimized a neural-network controller using elitism, roulette-wheel selection, and two-point crossover.

Additional Experience

VIP Research Team (BTAP) — Georgia Institute of Technology

Built an iOS AAC app for users with TBI/aphasia, implementing Visual Scene Display, text-to-speech, AI smart suggestions, and data persistence in Swift/SwiftUI.

Jan.–Dec. 2025

UN Peacekeeper — United Nations Interim Force in Lebanon

Maintained internal websites, servers, and computing devices for battalion communications; supported civil–military operations with multinational units.

Jun. 2023–Jan. 2024

Technical Skills

Programming: Python, C/C++, Java, JavaScript/TypeScript, MATLAB, SQL, R, Swift

Machine Learning & Generative AI: PyTorch, TensorFlow/Keras, scikit-learn, Transformers/ViTs, BERT/BioBERT, PEFT/LoRA, VAEs, Flow Matching, Hugging Face, OpenCV

Robotics: ROS 2, Franka Research 3, motion planning, impedance/admittance control, vision-guided manipulation

Data & Analytics: NumPy, Pandas, FAISS, statistical modeling, Tableau, Power BI

Systems & Deployment: Linux, Git, Docker, AWS, CUDA, ONNX, FastAPI, Streamlit, Jupyter/Colab, MLOps